



Sustainability and AI

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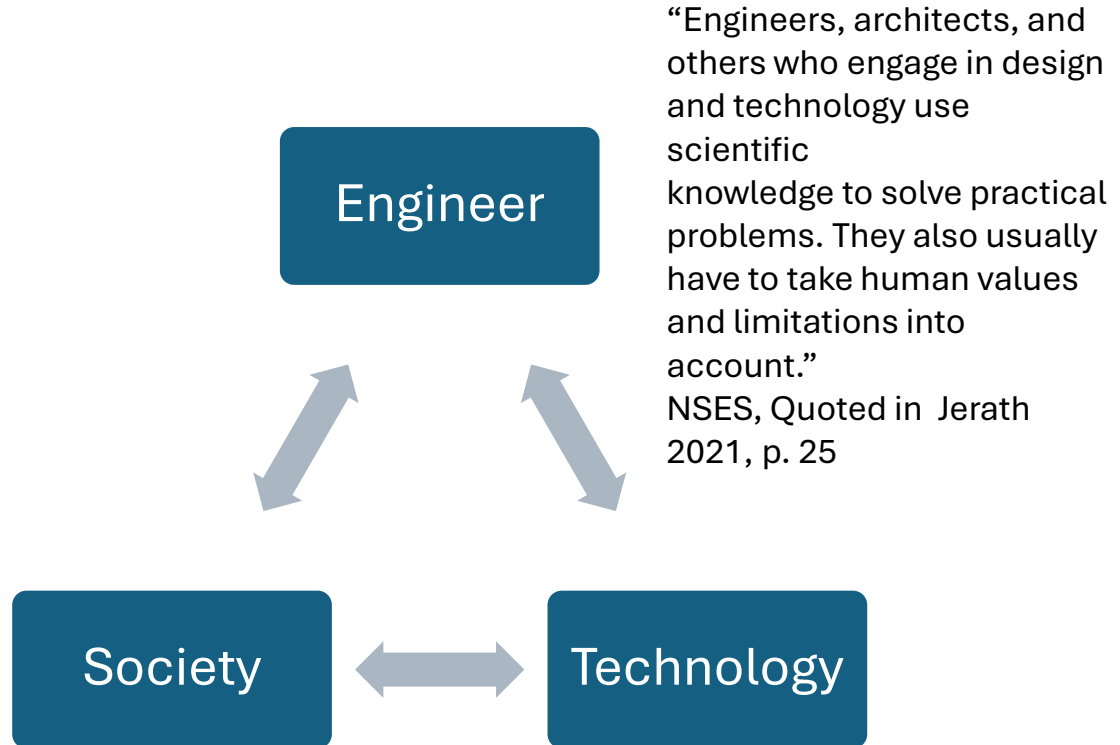
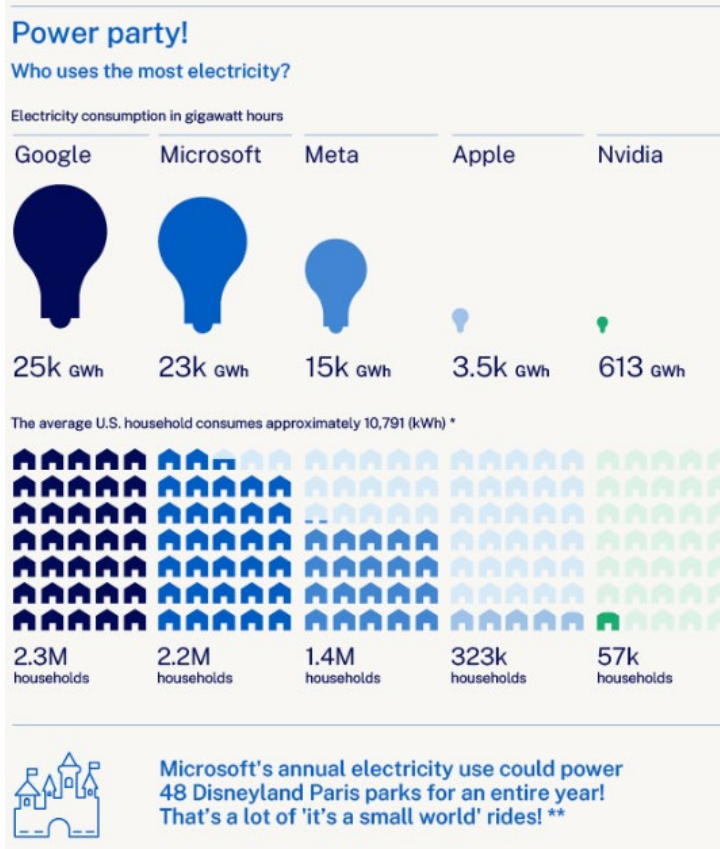
Key messages of the lecture

1. Sustainability is broader than just the environment, and AI has sustainability implications across many domains.
2. Achieving sustainability requires governance.
3. There are significant normative uncertainties about sustainability and AI, including “long-termism”.

Background: Introduction to sustainability and engineering

- Why sustainability matters
- Two perspectives on ethics and engineering
- Design for values

Why sustainability matters



1. Broadening our view of the impacts of AI

Narrow conception of technology

The functional “stuff” under the control of engineers and designers, that goes on to have a life in a real-life environment.

Broad conception of technology

That stuff, plus all the users, institutions, and social elements needed in order to make it do something in the real-world environment.

Van Wynsberghe broadens this to include the network of resources, infrastructure, and waste required to support a given technology.

Sociotechnical systems

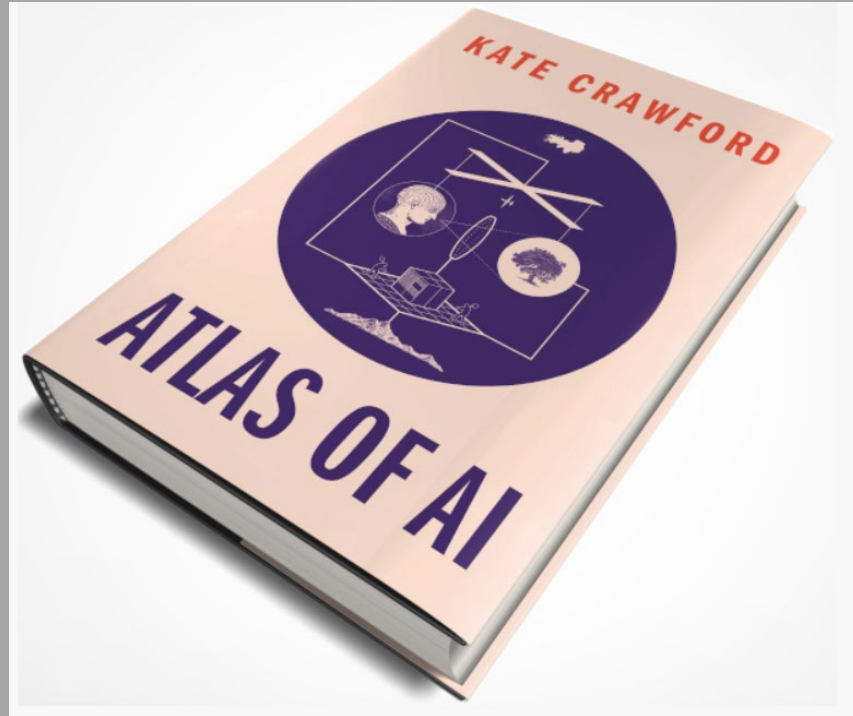
“Sociotechnical systems ... are hybrid systems, consisting of, or involving, “components” or “elements” that, by the description we have available for their behavior, belong to other domains than just the domain of the material objects --- described by natural science. Among these components or elements we may distinguish individual people, but also corporate actors such as business companies and governmental bodies, and more abstract social entities such as institutions, and laws, regulations and other rules.”

Franssen & Kroes 2009

Geography of AI (taken in the broadest sense)



Atlas of AI/ Anatomy of AI (Crawford): Anatomy of an AI System



Some elements of AI's sociotechnical system

Data servers

Mines for rare elements

Micro-workers

Some elements of AI's sociotechnical system

Data servers

Mines for rare elements

Micro-workers

“the production of AI is a labor-intensive process, which particularly needs the little-qualified, inconspicuous and low-paid contribution of “micro-workers” who annotate, tag, label, correct and sort the data that help to train and test smart solutions”

Tubaro & Casilli 2019

“Negative Externalities”

Externalities are those costs, risks, or harms created by an enterprise that do not appear on its balance sheet (hence they are imposed on others).

Examples are when hosting an AI model used by millions of people causes...

- Major disruptions to educational institutions
- Uncompensated pollution such as discard of old devices
- Strain on the energy grid

Without insuring or compensating for them.

We can use this idea to formulate a normative premise: Enterprises should insure or compensate for their negative externalities.

2. Meaning and implications of sustainability

The past meaning in English has concerned the acceptability of long-term environmental risks and harms;

Van Wynsberghe and others have begun to take a holistic view of sustainability, including social and political sustainability.

Environmental Values and the definition of sustainability

Brundtland: “development that meets the needs of the present without compromising the ability of future generations to meet their own needs”

[SDG Actions Platform | Department of Economic and Social Affairs](#)

Debates over the definition of sustainability

Sustainability as a concept has expanded to cover social, economic, and environmental aspects of a long-term perspective.

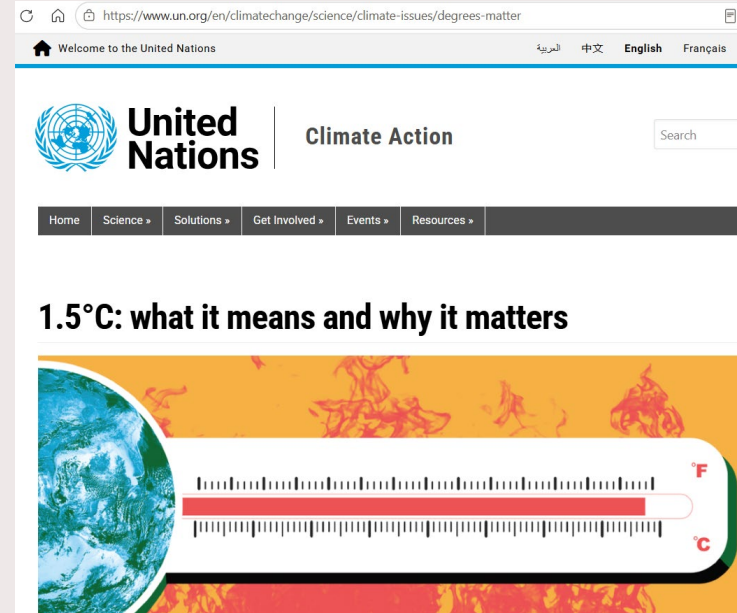
Debates over the definition of sustainability

Sustainability as a concept has expanded to cover social, economic, and environmental aspects of a long-term perspective.

Some have objected to this conceptual expansion, on the grounds that it “(a) obscures the real contradiction between the aims of welfare for all and environmental conservation; (b) risks diminishing the importance of the environmental dimension; and (c) separates social from economic aspects, which in reality are one and the same” (Kuhlman & Farrington 2010)

On the narrow definition: a reasonable normative premise (Lin 2022)

“Limiting global warming to 1.5°C and bringing greenhouse gas emissions to zero by the middle of the twenty-first century are necessary steps for mitigating the harmful effects of climate change on ecosystems and humanity’s wellbeing.”



Verduurzaming stagneert

Als de regering niet met extra maatregelen komt, ziet het PBL de plannen voor een groene samenleving somber in en blijft Nederland nog lang afhankelijk van import van fossiele energie.

Rosa Uijtewaal

DEN HAAG

Het zit niet mee met verduurzamen, maar dat Nederland zijn eigen klimaatdoelen niet haalt komt grotendeels door politieke keuzes. Dat was niet onvermijdelijk, benadrukt het Planbureau voor de Leefomgeving (PBL) in de jaarlijkse doorrekening van het Nederlandse klimaatbeleid.

Wie kijkt naar het klimaatdoel van 2030 (55 procent minder uitstoot van broeikasgassen dan in 1990) ziet dat Nederland er niet beter voor staat dan drie jaar geleden.

Opeenvolgende kabinetten hebben daar steken laten vallen, blijkt uit de doorrekening van het PBL. Zo zijn na de val van het kabinet-Rutte IV plannen controversieel verklaard en presenteerde het kabinet-Schoof maatregelen die klimaatdoelen verder weg brach-

ten. Ook het coalitieakkoord van de regering-Jetten doet weinig om Nederland op koers te brengen voor 2030. „Politieke impasses hebben het tempo uit de verduurzaming gehaald”, aldus de rekenmeesters van het planbureau.

In 2027 beslissen

Over eventuele extra klimaatplannen wil dit kabinet in het voorjaar van 2027 beslissen. Er is dan een „heel groot” pakket aan maatregelen nodig om de verduurzaming weer op gang te krijgen, zegt PBL-directeur Marko Hekkert. „Anders zien wij de transitie naar een klimaatneutrale samenleving niet gebeuren.” Dat het verduurzamen is stilgevallen noemt Hekkert „heel erg zorgelijk”.

De internationale afspraak uit het Klimaatakkoord van Parijs om de opwarming van de aarde tot onder 1,5 graden Celsius te houden, dreigt „buiten bereik” te raken,

schrijft het PBL. Het zou nog mogelijk zijn om de temperatuurstijging te beperken tot 2 graden (een cruciale grens om ernstige klimaatverandering te beperken), maar dan moet de wereld véél minder broeikasgassen uitstoten.

Nederland staat er niet beter voor dan drie jaar geleden

Op het ministerie van Economische Zaken en Klimaat zei minister Stientje van Veldhoven (Klimaat en Groene Groei, D66) maandag dat ze het ‘2030’-doel niet heeft „opgegeven”, maar dat „de kans dat we het halen heel erg klein is”. Volgens de minister was deze zomer vol hitte en droogtere records „een voorproefje” van de ernstige gevolgen van klimaatverandering en is het

nodig dat Nederland „klimaatverandering samen met andere landen probeert te beperken”.

Het kabinet sprak in het coalitieakkoord af „vol in” te zetten op plannen voor de langere termijn. De focus verschuift daarmee steeds meer naar 2040. Europese lidstaten hebben afgesproken in dat jaar 90 procent minder broeikasgassen uit te stoten dan in 1990. Met deze deadline in gedachten sprak de regering in het coalitieakkoord af miljarden uit te trekken voor een belangrijke verduurzamingssubsidie en voor windparken op zee.

Die maatregelen hebben pas effect ná 2030. Als alle plannen van het kabinet goed uitpakken, zou Nederland in 2040 ergens tussen de 61 en 72 procent uitkomen. Het is onzeker of de industrie vol-

Lees verder op pagina 3



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On the broad definition: AI could contribute to sustainability, but it also has a large impact on sustainability (CHLEP Ch. 6)

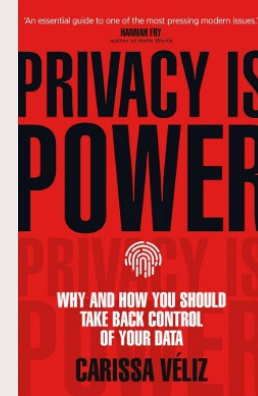
AI for sustainability: e.g., in relation to SDGs

- “Environmentally Sound Technologies”
- “Sustainable AI”

Sustainability of AI:

- Hasselbalch & Van Wynsberghe 2025: focused on the broad socio-technical system
- “data pollution” encompasses tangible and intangible impacts: not just environmental damage, but also data “spills,” privacy violations, discrimination.
- “Privacy ... [is] the needle on the gauge of society’s power balance” (Hasselbalch & Tranberg 2016)

Compare: <https://jolt.law.harvard.edu/digest/model-collapse-and-the-right-to-uncontaminated-human-generated-data>



“Power dynamics” and sustainability ethics at three levels in AI (CHLEP Ch. 6)

Micro: An engineer or a team of engineers makes choices about development of an AI tool, e.g., whether to make it depend on a big tech provider, how much computation to use. (individual actions)

Meso: Companies and institutions establish standards and policy. (setting patterns for large actors)

Macro: Global entities or the system as a whole reacts to and influences the historical and political trajectory of “system technologies” like AI. (setting the rules and the context for large actors)

Meso-level voluntary proposals

<https://www.sustainableaicoalition.org/> : “Shared Vision” and commitment to goals

First, they reiterate the AI Sustainability Dilemma.

Then, they pledge certain goals. (Next slides)

1. **Encourage AI Initiatives for the Planet:** We commend inclusive and collaborative efforts to harness AI for the benefit of the local and global environment, including its role in decarbonizing economies, reducing pollution, preserving biodiversity, protecting the oceans, and ensuring humanity operates within planetary boundaries.
2. **Wish to continue our work for an AI respectful of our Planetary Boundaries:** To this end, we understand the need to focus on the following:
 - **Standardized methods and metrics for measuring AI's environmental impacts:** We advocate for the development of consistent methods to better assess AI's environmental impacts, notably on energy consumption.
 - **Comprehensive Life Cycle Analysis:** AI systems should be evaluated across their entire life cycle through multi-criteria analysis. For software, this includes data collection, model development, training, deployment, and maintenance. For hardware, this involves mining and extraction practices, transportation, energy and water use, and waste management, including e-waste generation.
 - **Frameworks for Reporting and Disclosure:** Mechanisms should be established for companies to transparently report the environmental impact of their AI products and services.
 - **Prioritization of Research on Sustainable AI:** The private sector and research community should prioritize developing AI technologies with optimized algorithms to reduce computational complexity and minimize data usage.
 - **Efforts that ensure AI infrastructure and software are built and maintained in line with global environmental commitments.**
3. **Support Collaborative Approaches:** We recognize the essential role of multi-stakeholder collaboration—bringing together governments, academia, civil society, and the private sector—to advance these objectives. By pooling our knowledge and

Match the following levels of Hasselbalch & van Wynsberghe's sustainability analysis with the appropriate examples.

Micro	Social movements form in many countries to stop the construction of data centers.
Meso	A facial recognition company takes seriously its obligation to respect international human rights when contracting on tasks for data micro-workers
Macro	An engineer chooses to limit computation when developing an AI tool.
	Meta takes care when establishing carbon offset projects, so as not to harm indigenous populations.
	An international treaty is agreed on standards to track the electricity consumption of data standards

Sustainability at TU/e

Sustainability at TU/e does not strongly emphasize AI.

Surprisingly, it is difficult to find any statement about AI and sustainability on the TU/e website.

Sustainability @ TU/e - this is how



At TU/e we practice Sustainability: it is one of the key [strategic priorities](#) of our university. This is done with the aid of our [Sustainability Office](#) and in collaboration with key stakeholders, such as the [GO Green Office](#), and other groups from all four dimensions of the university. The Sustainability Office collaborates closely with our [Sustainability Ambassador, Anna J. Wieczorek](#) and in partnership with various internal and external departments and organizations. An Advisory Board for

Sustainability provides guidance and advice.

We are working on integrating Sustainability into the four dimensions of our university:

- [Research](#)
- [Education](#)
- [Campus Operations](#)
- [Governance](#)

Image from [AI and Sustainability: Opportunities, Challenges, and Impact | EY - Netherlands](#)



IEEE: Alignment at the meso-level for “autonomous and intelligent systems”(A/IS)

UNIVERSAL DECLARATION OF HUMAN RIGHTS -it's our Declaration



A/IS regulation, development, and deployment should, therefore, be based on international human rights standards and standards of international humanitarian laws (in the case of armed conflicts). This can be achieved if both states and private actors consider their responsibility to respectively protect and respect internationally recognized political, social, economic, and cultural rights. For business actors, this means considering their obligation to respect international human rights, as laid out in the UN Guiding Principles for Business and Human Rights (OHCHR, 2011), also known as the [Ruggie principles](#).

Three regulatory proposals in Lin 2022

1. Monitoring emissions impact (control → measurement)
2. Climate-aware guidelines for AI training
3. Clean energy for data centers

Omnibus passed in Dec 2025

The Great EU Reversal: Fast-Track Deregulation and the Erosion of Europe's Sustainability Ambitions Regulation

Behind the language of simplification, the EU's 2025 omnibus sustainability law, hollowing out the Green Deal, and setting back climate goals

by Sara Todeschini



Is AI policy special?

→ European data sovereignty, security, and industrial competitiveness via AI are so important that cuts in the carbon footprint should come elsewhere.

→ Lin's policy measures are too weak to reduce emissions substantially.

→ Lin's policy measures do not ensure elimination of relevant externalities.



NOS Nieuws • Vrijdag 10 juni 2022, 19:59

Hyperscale-datacenters in Nederland? Nog maar op twee plekken welkom

The footprint of machine learning



Three points of impact (Lin 2022)

1. The impact of training machine learning
2. The impact of building and maintaining data centers
3. Impact of real-life application of machine learning

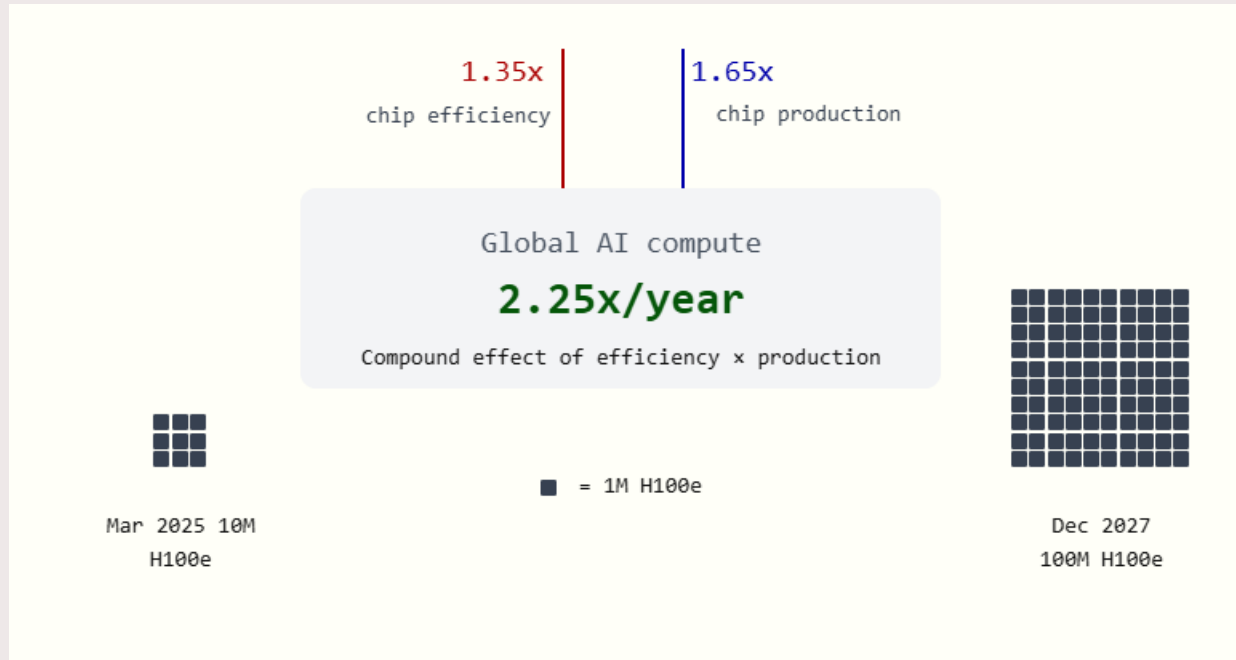
Measurement of the footprint

Some open-source resources can be used to estimate the impacts of LLMs on the environment, considering four factors:

- Training duration (amount of computation)
- Energy efficiency of GPUs
- Location of data center
- Offsets (if any)

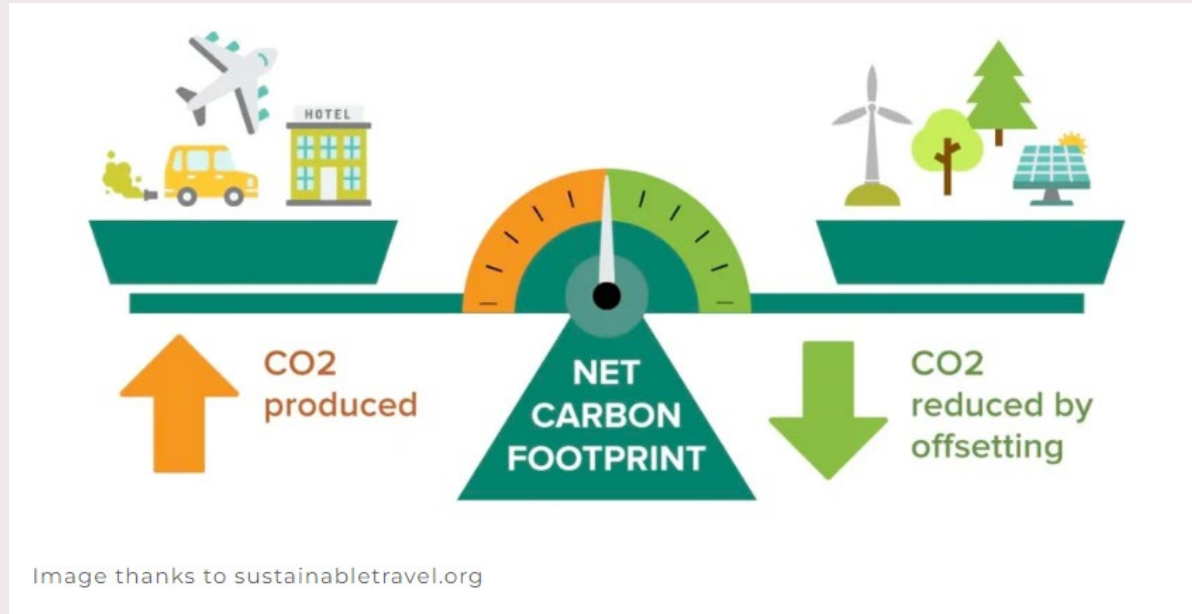
[AI and Sustainability: Opportunities, Challenges, and Impact | EY - Netherlands](#)

Estimating the future computing needed for training AI



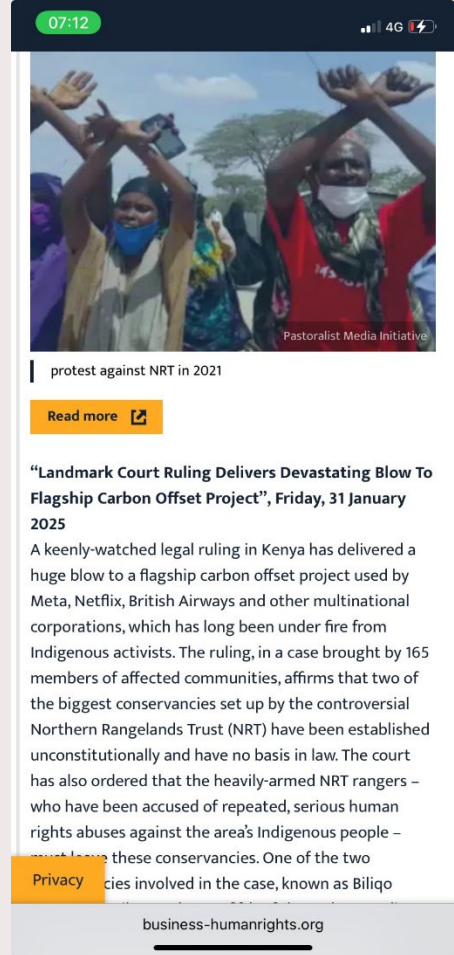
Kokotajlo et al 2025, <https://ai-2027.com/research/compute-forecast>

Offsetting



Offsetting can be more problematic than expected

Alvarez, C. (1 Feb 2025) Kenya: Court halts flagship carbon offset project used by Meta, Netflix, and British Airways over unlawfully acquiring community land without consent, accessed at business-humanrights.org on 18 September 2025



3. Normative Uncertainties around Sustainable AI

Individual use of AI tools

People sometimes worry about the impacts of individual use of AI.

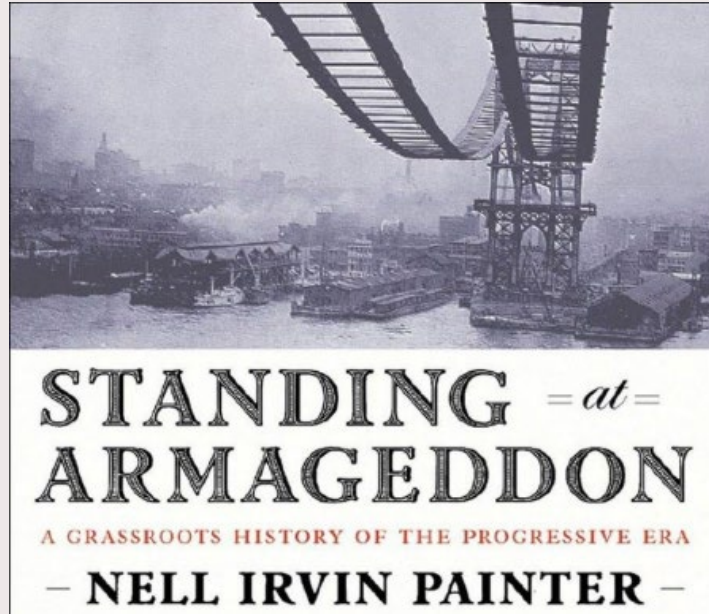
Andy Masley has argued that the impact of individual use is small enough as to be morally negligible. According to his (plausible) calculations, "It takes ~1000 prompts to raise your emissions by 1%."

<https://www.andymasley.com/writing/a-short-summary-of-my-argument-that/>

Do you think this is the basis of a good counterargument?



Sustainability and the historical moment



Very long term effects of AI

“Long-termism” is the view that we should strongly weight the distant future effects of AI on future generations. This includes both threats (AGI takes over the world, somebody uses AI to develop superweapons) and prospects (we solve all human problems with AI).

In addition to uncertainty about these effects, long-termism is itself morally controversial.

Key messages of the lecture

1. Sustainability is broader than just the environment, and AI has sustainability implications across many domains.
2. Achieving sustainability requires governance.
3. There are significant normative uncertainties about sustainability and AI, including “long-termism”.

Literature

<https://www.sustainableaicoalition.org/coalition/>

Hasselbalch & van Wynsberghe. Artificial Intelligence, Power, and Sustainability. In Smuha, N. A., ed. (2025). *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence*. Cambridge University Press. Pp. 117-130.

Lin, Patrick K. (2022). The Cost of Teaching a Machine: Lighting the Way for a Climate-Aware Policy Framework That Addresses Artificial Intelligence's Carbon Footprint Problem. *Fordham Env't L. Rev.* 34.